

# Senior QA Assignment

## Objective

This assignment is designed to evaluate your ability to think beyond UI testing and demonstrate:

- QA and system-level thinking
- Risk-based testing
- API/backend understanding
- Automation strategy and framework thinking
- Ability to make practical testing decisions

**Expected effort:** 2–3 hours

Please focus on the **quality of your thinking rather than the quantity of test cases or code.**

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## Scenario

You are testing a **Retail Audit Platform** used by field employees.

The product consists of:

- Android & iOS mobile applications
- Web application
- REST APIs
- Backend services
- Cloud object storage
- Asynchronous processing

A typical workflow is:

**Login → Open Camera → Capture Photos → Upload Photos to Backend → Backend Processing → AI Results → Web Dashboard**

A user can capture anywhere between **10 and 100 photos** in a session.

The application needs to work reliably even when users have:

- Poor/intermittent network connectivity
  - Wi-Fi/mobile network changes
  - Application backgrounding or termination
  - Different mobile devices and OS versions
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## Production Issue

A customer reports:

"I captured 20 photos. The mobile application shows all 20 photos as uploaded. However, the web dashboard shows only 18 processed photos. Restarting the mobile application sometimes makes the missing photos appear."

The engineering team provides the following information:

### Mobile logs

```
1 | PHOTO-101 → Upload completed
2 | PHOTO-102 → Upload completed
3 | PHOTO-103 → Upload completed
```

### Backend

```
1 | PHOTO-101 → UPLOADED
2 | PHOTO-102 → UPLOADED
3 | PHOTO-103 → PROCESSING
```

### Cloud Storage

```
1 | PHOTO-101 → Exists
2 | PHOTO-102 → Exists
3 | PHOTO-103 → NOT FOUND
```

### Web

```
1 | Total Photos: 20
2 | Processed: 18
```

## Assignment

### 1. Investigation & QA Thinking

Based on the above scenario, answer the following:

A. What are the 5 most likely failure points?

Explain your reasoning.

B. What additional information would you request?

Consider:

- Application logs
- API logs

- Backend logs
- Database
- Cloud storage
- Events/queues
- Monitoring/metrics
- Request/correlation IDs

C. How would you reproduce this issue?

Mention the key scenarios you would try.

For example:

- Network interruption
- Wi-Fi → mobile data
- Mobile data → Wi-Fi
- App background/termination
- Upload timeout
- API timeout
- Retry
- App restart

You do not need to provide exhaustive test cases.

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## 2. Test Strategy

Create **10 high-value test scenarios** for this workflow.

Your scenarios should cover more than just UI testing.

For each scenario, mention:

| Scenario                                            | Test Layer | Expected Result                                    | Automation |
|-----------------------------------------------------|------------|----------------------------------------------------|------------|
| Example:<br>Network disconnects during photo upload | API/Mobile | Upload eventually succeeds without duplicate photo | Yes        |

Consider different layers such as:

- API
- Backend/service
- Mobile
- Web
- End-to-End

#### Important

We are interested in **why you chose a particular test layer**.

For example:

"I would test this at API level instead of UI level because..."

### 3. Automation Architecture

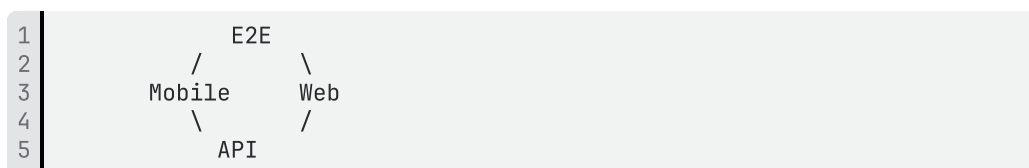
Assume you join the team and discover that most existing automation is UI-based and takes **3 hours** to execute.

Design a high-level automation strategy for the product.

Show where you would use:

- API tests
- Mobile UI tests
- Web UI tests
- End-to-End tests

You may provide a simple diagram such as:



However, design the architecture that you believe is appropriate.

Explain briefly:

1. Which tests should **not** be UI tests?
2. How would you manage test data?
3. How would you handle flaky tests?
4. How would you integrate the automation into CI/CD?

**Maximum: 1 page.**

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## 4. API Automation

Use the publicly available **Restful-Booker API** for this part.

### API Documentation:

<https://www.postman.com/automation-in-testing/restful-booker-collections/documentation/55eh7vh/restful-booker>

### API:

[Welcome to Restful-Booker](#)

Automate the following workflow:

**Authenticate → Create Booking → Get Booking → Update Booking → Delete Booking**

Create **5–8 meaningful automated tests**.

Your tests should demonstrate:

- Positive testing
- Negative testing
- Meaningful assertions
- Dynamic test data
- Test independence
- Reusable code
- Maintainable framework structure

Do not attempt to automate every available endpoint.

We are more interested in **which tests you choose and why**.

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## Deliverables

Please submit:

### 1. Investigation & QA Thinking

Maximum **2 pages**

### 2. Test Scenarios

10 high-value scenarios

### 3. Automation Architecture

Maximum **1 page**

#### 4. API Automation

Source code + a short README explaining:

- Prerequisites
  - How to run the tests
  - Framework/language used
  - Any assumptions
- 

### Evaluation Criteria

| Area                           | Weight      |
|--------------------------------|-------------|
| QA / System Thinking           | 30%         |
| Risk-based Test Design         | 20%         |
| Automation Architecture        | 20%         |
| API Automation                 | 20%         |
| Code Quality & Maintainability | 10%         |
| <b>Total</b>                   | <b>100%</b> |

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### What We Are Looking For

We are **not** looking for the person who produces the largest number of test cases.

We are looking for someone who can:

- Identify the most important risks
- Think beyond UI testing
- Understand APIs and backend workflows
- Reason about failures and retries
- Identify data consistency problems
- Design an appropriate automation strategy
- Choose the right testing layer
- Build maintainable automation
- Explain technical trade-offs
- Think about quality as an engineering responsibility

If a requirement is ambiguous, **state your assumption** rather than trying to guess the intended behavior.

There is no single "correct" solution. We will evaluate the **reasoning behind your decisions**.

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### Expected Time

**2–3 hours**

Please do not spend excessive time making the document visually polished. We are primarily evaluating your **technical thinking, QA mindset, and automation approach**.